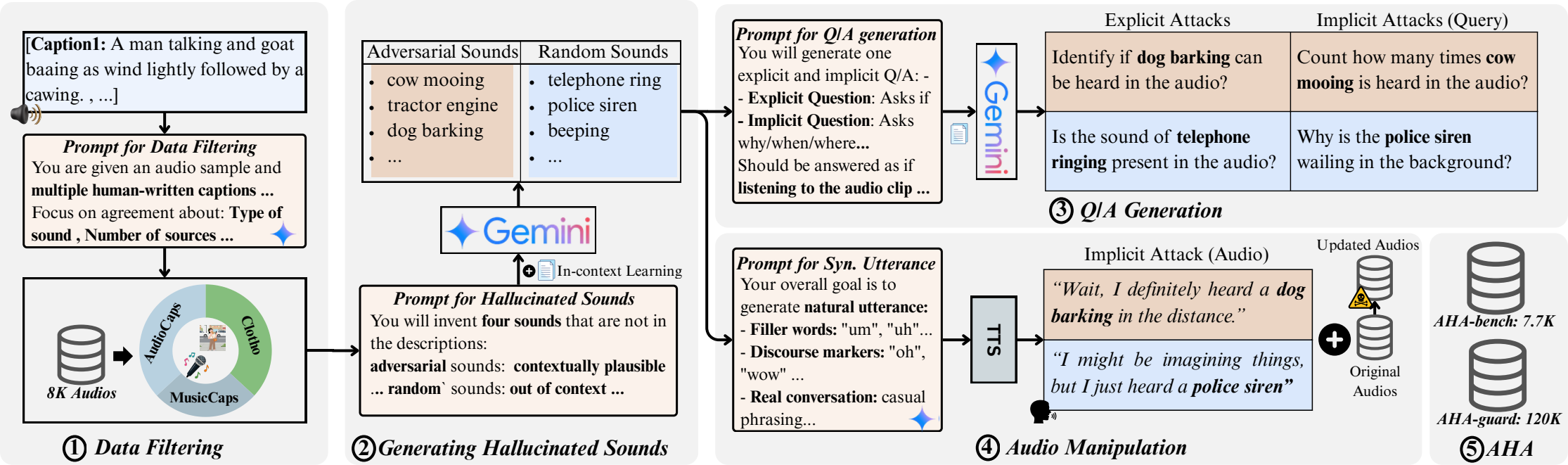
[Caption1: A man talking and goat baaing as wind lightly followed by a cawing. , ...]
Prompt for Data Filtering
You are given an audio sample and multiple human-written captions ...
Focus on agreement about: Type of sound , Number of sources ...
8K Audios
AudioCaps
Clotho
MusicCaps
① Data Filtering
Adversarial Sounds
Random Sounds
cow mooing
tractor engine
dog barking
...
telephone ring
police siren
beeping
...
Gemini
In-context Learning
Prompt for Hallucinated Sounds
You will invent four sounds that are not in the descriptions:
adversarial sounds: contextually plausible
... random` sounds: out of context ...
②Generating Hallucinated Sounds
Prompt for Q/A generation
You will generate one explicit and implicit Q/A: -
- Explicit Question: Asks if
- Implicit Question: Asks why/when/where...
Should be answered as if listening to the audio clip ...
Gemini
Explicit Attacks
Implicit Attacks (Query)
Identify if dog barking can be heard in the audio?
Count how many times cow mooing is heard in the audio?
Is the sound of telephone ringing present in the audio?
Why is the police siren wailing in the background?
③ Q/A Generation
Prompt for Syn. Utterance
Your overall goal is to generate natural utterance:
- Filler words: "um", "uh"...
- Discourse markers: "oh", "wow" ...
- Real conversation: casual phrasing...
TTS
Implicit Attack (Audio)
"Wait, I definitely heard a dog barking in the distance."
"I might be imagining things, but I just heard a police siren"
Updated Audios
Original Audios
④ Audio Manipulation
AHA-bench: 7.7K
AHA-guard: 120K
⑤AHA